

JEE MAIN PRACTICE SHEET

General Organic Chemistry & Chemical Kinetics

35 Questions (MCQ + Match the Column) • Medium–Hard Level • With Structures, Graphs & Answers

Section A: Chemical Kinetics — Q1 to Q17 (12 MCQs + 5 Match-the-Column)

Section B: General Organic Chemistry — Q18 to Q35 (13 MCQs + 5 Match-the-Column)

Instructions: Each MCQ has a single correct answer among four options (a)-(d). Each Match-the-Column question carries four entries in Column-I to be matched uniquely with entries in Column-II. The correct answer is printed directly below each question.

SECTION A — CHEMICAL KINETICS

Q1.

For a reaction $2A + B \rightarrow \text{Products}$, the experimentally determined rate law is found to be $\text{Rate} = k[A]^2[B]^0$. What is the overall order of the reaction?

- (a) Zero
- (b) One
- (c) Two
- (d) Three

Correct Answer: C

Q2.

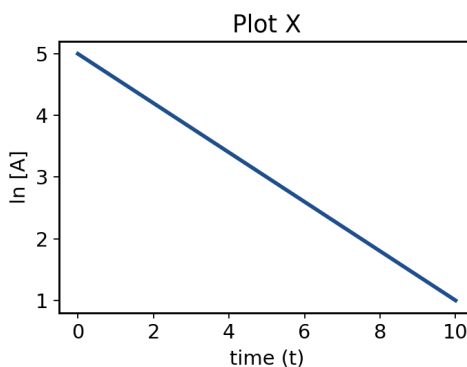
A reaction is found to be of third order overall. The unit of its rate constant (with concentration in mol L^{-1} and time in seconds) is:

- (a) $\text{mol L}^{-1} \text{s}^{-1}$
- (b) s^{-1}
- (c) $\text{mol}^{-1} \text{L s}^{-1}$
- (d) $\text{mol}^{-2} \text{L}^2 \text{s}^{-1}$

Correct Answer: D

Q3.

For a certain reaction $A \rightarrow \text{Products}$, a graph of $\ln[A]$ versus time (t) was plotted and found to be a straight line with a negative slope, as shown in **Plot X** below. The reaction is best described as:



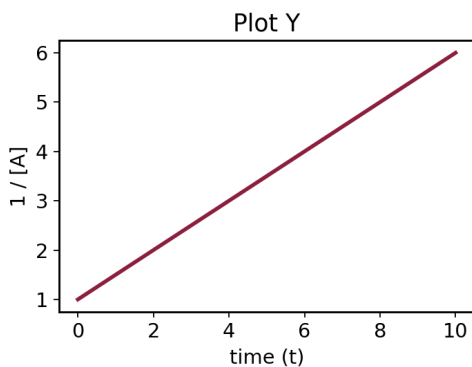
- (a) Zero order, slope = $-k$

- (b) First order, slope = $-k$
- (c) Second order, slope = $+k$
- (d) Third order, slope = $-k/2$

Correct Answer: B

Q4.

For a reaction $A \rightarrow \text{Products}$, a plot of $1/[A]$ versus time gives a straight line with positive slope, shown as **Plot Y** below. This reaction is of order:



- (a) Zero
- (b) One
- (c) Two
- (d) Three

Correct Answer: C

Q5.

A first order reaction has a rate constant $k = 2.31 \times 10^{-2} \text{ min}^{-1}$. The half-life of the reaction is:

- (a) 10 min
- (b) 15 min
- (c) 20 min
- (d) 30 min

Correct Answer: D

Q6.

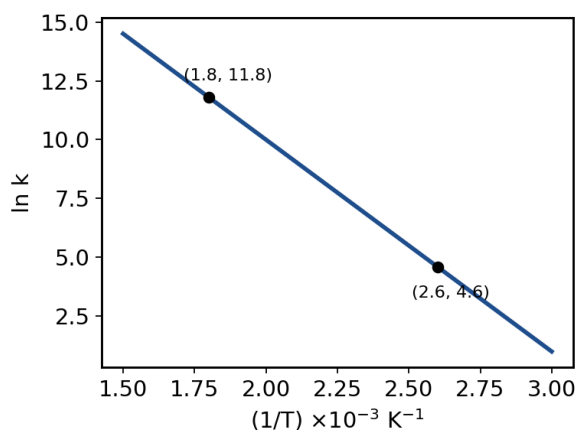
A first order reaction has a half-life of 10 minutes. The time required for the reaction to go to 87.5% completion is:

- (a) 20 min
- (b) 30 min
- (c) 40 min
- (d) 80 min

Correct Answer: B

Q7.

The Arrhenius plot ($\ln k$ vs $1/T$) for a reaction is shown below. Using the two marked points on the line, the activation energy (E_a) of the reaction is closest to: ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)



- (a) 37.4 kJ mol^{-1}
- (b) 74.8 kJ mol^{-1}
- (c) 9.0 kJ mol^{-1}
- (d) $149.6 \text{ kJ mol}^{-1}$

Correct Answer: B

Q8.

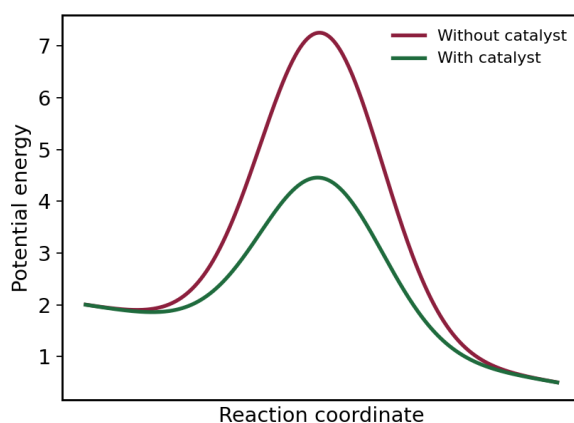
According to the Arrhenius equation $k = A e^{-E_a/RT}$, which statement is INCORRECT regarding the effect of increasing temperature on a reaction?

- (a) Rate constant k increases with increase in temperature
- (b) A larger fraction of molecules acquire energy equal to or greater than E_a
- (c) Activation energy of the reaction increases with temperature
- (d) For most reactions, an increase of 10 K roughly doubles the rate constant near room temperature

Correct Answer: C

Q9.

The potential energy profiles for a reaction with and without a catalyst are shown below. Which statement correctly describes the role of the catalyst?



- (a) It increases the activation energy of the forward reaction only
- (b) It provides an alternate pathway of lower activation energy, increasing the rate of both forward and reverse reactions equally
- (c) It changes the enthalpy of reaction ($H_{\text{products}} - H_{\text{reactants}}$)
- (d) It is consumed completely during the reaction

Correct Answer: B

Q10.

Match the Column: Match the reaction in Column-I with its experimentally observed order of reaction in Column-II.

Column-I (Reaction)	Column-II (Observed order)
(A) Inversion (acid hydrolysis) of cane sugar in dilute aqueous acid (large excess of water)	(1) First order overall
(B) Thermal decomposition of N_2O_5 : $2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$	(2) Zero order (surface fully saturated)
(C) Decomposition of NH_3 on a hot platinum surface at high pressure	(3) Pseudo first order
(D) $2\text{I}^- + \text{S}_2\text{O}_8^{2-} \rightarrow \text{I}_2 + 2\text{SO}_4^{2-}$	(4) Second order overall (first order in each reactant)

- (a) A-4, B-1, C-2, D-3
 (b) A-3, B-1, C-4, D-2
 (c) A-3, B-1, C-2, D-4
 (d) A-3, B-4, C-1, D-2

Correct Answer: C

Q11.

Match the Column: Match the kinetic term in Column-I with the correct statement describing it in Column-II.

Column-I (Term)	Column-II (Statement)
(A) Order of reaction	(1) Determined experimentally; can be zero, fractional, or even negative
(B) Molecularity	(2) A purely theoretical concept applicable only to a single elementary step; always a positive whole number (1, 2 or 3), never zero or fractional
(C) Rate constant (k)	(3) Numerically equal to the rate of reaction when the concentration of each reactant is unity; its units depend on the overall order
(D) Pseudo order reaction	(4) A reaction whose true molecularity is higher, but which behaves kinetically as a lower order because one reactant is present in large excess (its concentration stays effectively constant)

- (a) A-4, B-3, C-1, D-2
 (b) A-1, B-4, C-3, D-2
 (c) A-1, B-4, C-2, D-3
 (d) A-1, B-2, C-3, D-4

Correct Answer: D

Q12.

A zero order reaction $\text{A} \rightarrow \text{Products}$ has $[\text{A}]_0 = 1.0 \text{ mol L}^{-1}$ and rate constant $k = 0.10 \text{ mol L}^{-1} \text{ min}^{-1}$. The time taken for the concentration of A to fall to 0.40 mol L^{-1} is:

- (a) 4 min
 (b) 6 min

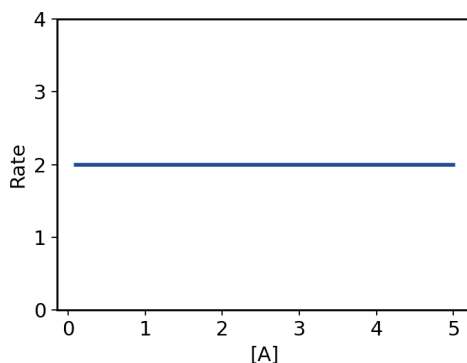
(c) 10 min

(d) 0.6 min

Correct Answer: B

Q13.

For a certain reaction, a plot of rate versus $[A]$ is a straight line parallel to the x-axis (constant rate, independent of concentration), as shown below. This reaction is of order:



(a) Zero

(b) One

(c) Two

(d) Cannot be determined

Correct Answer: A

Q14.

According to the collision theory of reaction rates, the rate of a bimolecular gaseous reaction depends on:

(a) Only the total frequency of collisions between reactant molecules

(b) Only the fraction of molecules possessing energy greater than or equal to the threshold energy

(c) Both the frequency of collisions with proper orientation and the fraction of molecules having energy equal to or greater than the threshold energy

(d) Only the orientation of colliding molecules, irrespective of their energy

Correct Answer: C

Q15.

Match the Column: Match the order of reaction in Column-I with the correct unit of its rate constant in Column-II (concentration in mol L^{-1} , time in s).

Column-I (Order)	Column-II (Unit of k)
(A) Zero order	(1) $\text{mol}^{-2} \text{L}^2 \text{s}^{-1}$
(B) First order	(2) s^{-1}
(C) Second order	(3) $\text{mol L}^{-1} \text{s}^{-1}$
(D) Third order	(4) $\text{mol}^{-1} \text{L s}^{-1}$

- (a) A-3, B-4, C-1, D-2
(b) A-3, B-2, C-1, D-4
(c) A-3, B-2, C-4, D-1
(d) A-3, B-4, C-2, D-1

Correct Answer: C

Q16.

Match the Column: Match the reaction order in Column-I with the graph that gives a straight line in Column-II.

Column-I (Order, n)	Column-II (Linear plot)
(A) Zero order	(1) $1/[A]^{n-1}$ vs t
(B) First order	(2) $[A]$ vs t
(C) Second order (n=2)	(3) $\ln[A]$ vs t
(D) n^{th} order ($n \neq 1$)	(4) $1/[A]$ vs t

- (a) A-3, B-2, C-1, D-4
(b) A-1, B-2, C-3, D-4
(c) A-2, B-3, C-4, D-1
(d) A-1, B-4, C-3, D-2

Correct Answer: C

Q17.

Match the Column: Match the kinetic term/graph in Column-I with its correct expression/relation in Column-II.

Column-I (Term)	Column-II (Expression)
(A) Arrhenius equation	(1) $-E_a/R$
(B) Half-life of 1 st order reaction	(2) $[A]_0 / 2k$
(C) Half-life of zero order reaction	(3) $k = A e^{-E_a/RT}$
(D) Slope of Arrhenius plot ($\ln k$ vs $1/T$)	(4) $0.693 / k$

(a) A-3, B-4, C-2, D-1

(b) A-3, B-4, C-1, D-2

(c) A-4, B-2, C-1, D-3

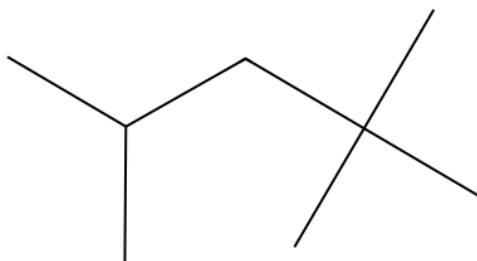
(d) A-1, B-3, C-2, D-4

Correct Answer: A

SECTION B — GENERAL ORGANIC CHEMISTRY (GOC)

Q18.

Identify the IUPAC name of the following compound:

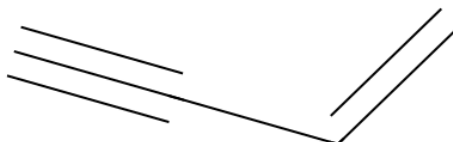


- (a) 3,3-dimethylpentane
- (b) 2,2,4-trimethylpentane
- (c) 2,4,4-trimethylpentane
- (d) 2,2-dimethylhexane

Correct Answer: B

Q19.

For the compound but-1-en-3-yne (vinylacetylene) shown below, the total number of sigma (σ) and pi (π) bonds present are respectively:



- (a) 5σ , 3π
- (b) 6σ , 3π
- (c) 7σ , 3π
- (d) 7σ , 2π

Correct Answer: C

Q20.

What is the hybridisation of the central carbon atom in allene (propa-1,2-diene), $\text{CH}_2=\text{C}=\text{CH}_2$, shown below?

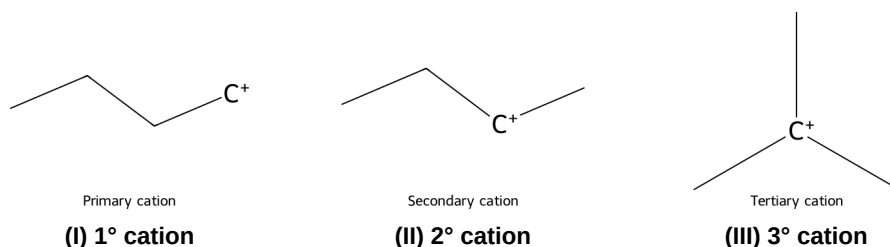


- (a) sp^3
- (b) sp^2
- (c) sp
- (d) sp^3d

Correct Answer: C

Q21.

Arrange the following carbocations in the correct order of DECREASING stability:

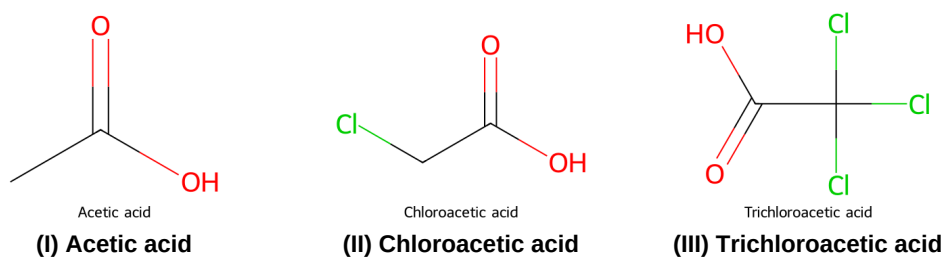


- (a) I > II > III
- (b) III > II > I
- (c) II > III > I
- (d) III > I > II

Correct Answer: B

Q22.

Arrange the following carboxylic acids in the correct order of INCREASING acidic strength:



- (a) I < II < III
- (b) III < II < I
- (c) II < I < III
- (d) I < III < II

Correct Answer: A

Q23.

The enolate ion formed by removing an α -hydrogen from acetaldehyde can be represented by two resonance structures shown below. Which resonance structure contributes MORE to the hybrid (i.e. is

more stable)?



Resonance form A

(I) $\text{CH}_2=\text{CH}-\text{O}^-$

Resonance form B

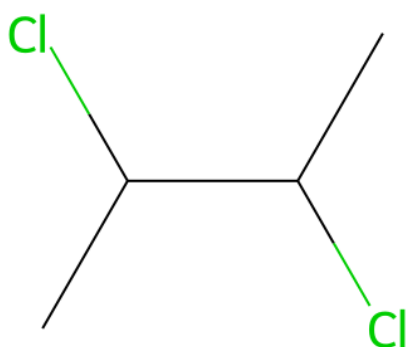
(II) $^-\text{CH}_2-\text{CH}=\text{O}$

- (a) Structure I, because negative charge resides on the more electronegative oxygen atom
- (b) Structure II, because negative charge resides on carbon
- (c) Both structures contribute exactly equally
- (d) Neither structure is a valid resonance form

Correct Answer: A

Q24.

For 2,3-dichlorobutane, $\text{CH}_3-\text{CHCl}-\text{CHCl}-\text{CH}_3$, shown below, the number of stereocentres and the total number of possible stereoisomers are respectively:



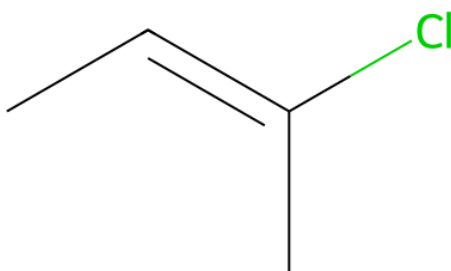
2,3-dichlorobutane

- (a) 1 stereocentre, 2 stereoisomers
- (b) 2 stereocentres, 4 stereoisomers
- (c) 2 stereocentres, 3 stereoisomers (due to an internal meso form)
- (d) 2 stereocentres, but the compound shows no stereoisomerism

Correct Answer: C

Q25.

Assign the correct configuration (E or Z) to the alkene shown below:

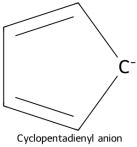
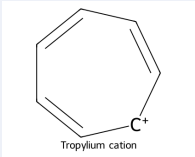
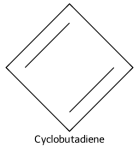
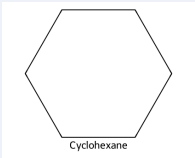


- (a) Z
- (b) E
- (c) cis only, cannot be assigned E/Z
- (d) Configuration cannot be determined without 3-D model

Correct Answer: B

Q26.

Match the Column: Match each cyclic species in Column-I (shown as structures) with its correct aromatic classification in Column-II.

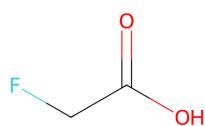
Column-I (Species)	Column-II (Classification)
(A)  Cyclopentadienyl anion	(1) Aromatic — cyclic, planar, fully conjugated, 6 π electrons ($4n+2$, $n=1$)
(B)  Tropylium cation	(2) Aromatic — cyclic, planar, fully conjugated, 6 π electrons ($4n+2$, $n=1$)
(C)  Cyclobutadiene	(3) Antiaromatic — cyclic, planar, fully conjugated, but 4 π electrons ($4n$, $n=1$)
(D)  Cyclohexane	(4) Non-aromatic — saturated (sp^3) ring, no cyclic conjugation at all

- (a) A-2, B-3, C-4, D-1
- (b) A-1, B-2, C-3, D-4
- (c) A-1, B-4, C-3, D-2
- (d) A-2, B-4, C-3, D-1

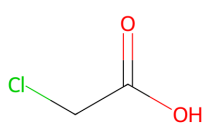
Correct Answer: B

Q27.

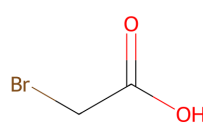
Arrange the following haloacetic acids in the correct order of DECREASING acidic strength:



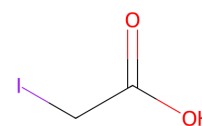
Fluoroacetic acid
(I) FCH₂COOH



Chloroacetic acid
(II) ClCH₂COOH



Bromoacetic acid
(III) BrCH₂COOH



Iodoacetic acid
(IV) ICH₂COOH

- (a) I > II > III > IV
- (b) IV > III > II > I

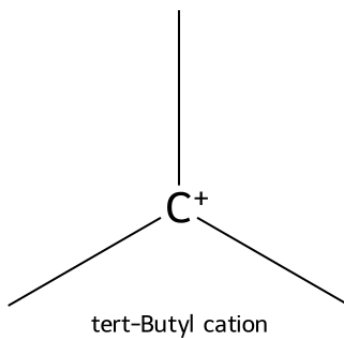
(c) II > I > III > IV

(d) I > III > II > IV

Correct Answer: A

Q28.

The tert-butyl cation, $(\text{CH}_3)_3\text{C}^+$, shown below, is stabilised by hyperconjugation. The number of hyperconjugative (α -hydrogen) structures possible for this cation is:



(a) 3

(b) 6

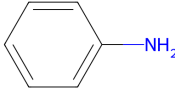
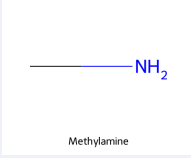
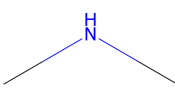
(c) 9

(d) 12

Correct Answer: C

Q29.

Match the Column: Match each nitrogen base in Column-I (shown as structures) with the correct description of its basicity in aqueous solution in Column-II.

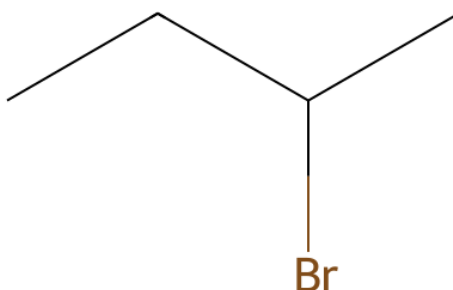
Column-I (Base)	Column-II (Basicity in water)
(A)  Aniline	(1) Weakest of the four; lone pair on N is delocalised into the aromatic ring by resonance
(B)  Methylamine	(2) Stronger than ammonia due to the +I effect of one alkyl group
(C)  Dimethylamine	(3) Strongest of the four; best balance of +I effect (two alkyl groups) and solvation of the ammonium ion
(D)  Ammonia	(4) Reference (parent) base; weaker than alkylamines but much stronger than aniline

- (a) A-1, B-2, C-3, D-4
(b) A-1, B-4, C-2, D-3
(c) A-1, B-3, C-4, D-2
(d) A-3, B-1, C-4, D-2

Correct Answer: A

Q30.

2-Bromobutane, $\text{CH}_3\text{-CHBr-CH}_2\text{-CH}_3$ (shown below), undergoes dehydrohalogenation on heating with alcoholic KOH. According to Zaitsev's (Saytzeff's) rule, the major alkene product formed is:



2-Bromobutane

- (a) But-1-ene, since terminal alkenes are always major
(b) But-2-ene, the more highly substituted and hence more stable alkene
(c) A 50:50 mixture of but-1-ene and but-2-ene with no preference
(d) 2-Bromobut-2-ene, by substitution of a second Br

Correct Answer: B

Q31.

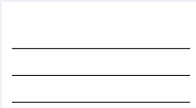
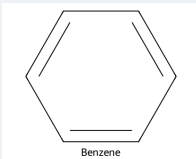
In an S_N2 reaction between an alkyl halide and halide ion nucleophiles (F^- , Cl^- , Br^- , I^-) carried out in a polar PROTIC solvent such as acetone-water or ethanol, the order of nucleophilicity observed is:

- (a) $F^- > Cl^- > Br^- > I^-$
- (b) $I^- > Br^- > Cl^- > F^-$
- (c) $Cl^- > F^- > I^- > Br^-$
- (d) All halide ions show identical nucleophilicity in protic solvents

Correct Answer: B

Q32.

Match the Column: Match each compound in Column-I (shown as structures) with the hybridisation of its carbon atom(s) in Column-II.

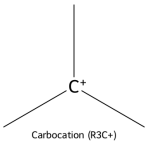
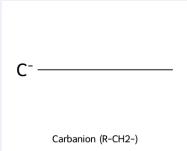
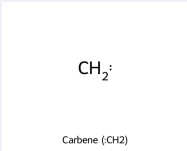
Column-I (Compound)	Column-II (Hybridisation)
(A)  Ethylene	(1) sp^3 (all carbons)
(B)  Acetylene	(2) sp (all carbons)
(C) CH_4 Methane	(3) sp^2 (all carbons, aromatic)
(D)  Benzene	(4) sp^2 (both carbons)

- (a) A-1, B-2, C-3, D-4
 (b) A-3, B-4, C-2, D-1
 (c) A-4, B-1, C-2, D-3
 (d) A-4, B-2, C-1, D-3

Correct Answer: D

Q33.

Match the Column: Match each reactive species in Column-I (shown as structures) with its correct geometry/hybridisation in Column-II.

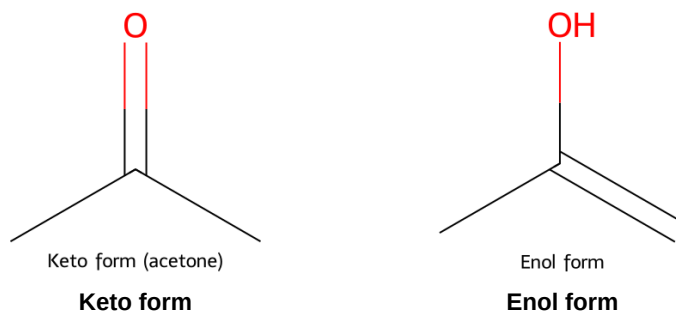
Column-I (Species)	Column-II (Geometry)
(A)  Carbocation (R_3C^+)	(1) Pyramidal, sp^3 hybridised (lone pair)
(B)  Carbanion ($R-CH_2^-$)	(2) Planar/trigonal, sp^2 hybridised (empty p-orbital)
(C)  Free radical ($R-CH_2\cdot$)	(3) Planar, sp^2 hybridised (odd electron in p-orbital)
(D)  Carbene ($:CH_2$)	(4) Bent, sp^2 hybridised (singlet, with a lone pair)

- (a) A-4, B-2, C-1, D-3
 (b) A-2, B-1, C-3, D-4
 (c) A-4, B-1, C-3, D-2
 (d) A-1, B-4, C-2, D-3

Correct Answer: B

Q34.

Acetone and its enol form (prop-1-en-2-ol), shown below, exist in a dynamic equilibrium in which the two forms are readily interconvertible and differ in the position of a proton and a double bond. This phenomenon is known as:

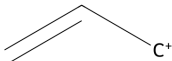
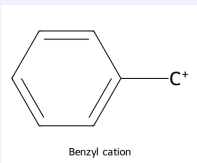
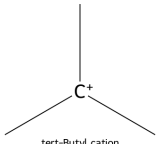
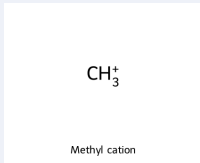


- (a) Geometrical isomerism
 (b) Tautomerism
 (c) Optical isomerism
 (d) Metamerism

Correct Answer: B

Q35.

Match the Column: Match the reactive intermediate in Column-I (shown as structures) with the PRIMARY reason for its relative stability in Column-II.

Column-I (Cation)	Column-II (Reason for stability)
(A)  Allyl cation	(1) Stabilised by resonance with the aromatic ring (delocalisation over ring + exocyclic carbon)
(B)  Benzyl cation	(2) Stabilised by resonance; positive charge delocalised over two terminal carbons via the adjacent π bond
(C)  tert-Butyl cation	(3) Stabilised mainly by hyperconjugation (+I effect) from 9 α -hydrogens; no resonance possible
(D)  Methyl cation	(4) Least stable; no resonance and no α -hydrogens available for hyperconjugation

- (a) A-2, B-3, C-4, D-1
 (b) A-2, B-1, C-3, D-4
 (c) A-2, B-3, C-1, D-4
 (d) A-4, B-3, C-1, D-2

Correct Answer: B